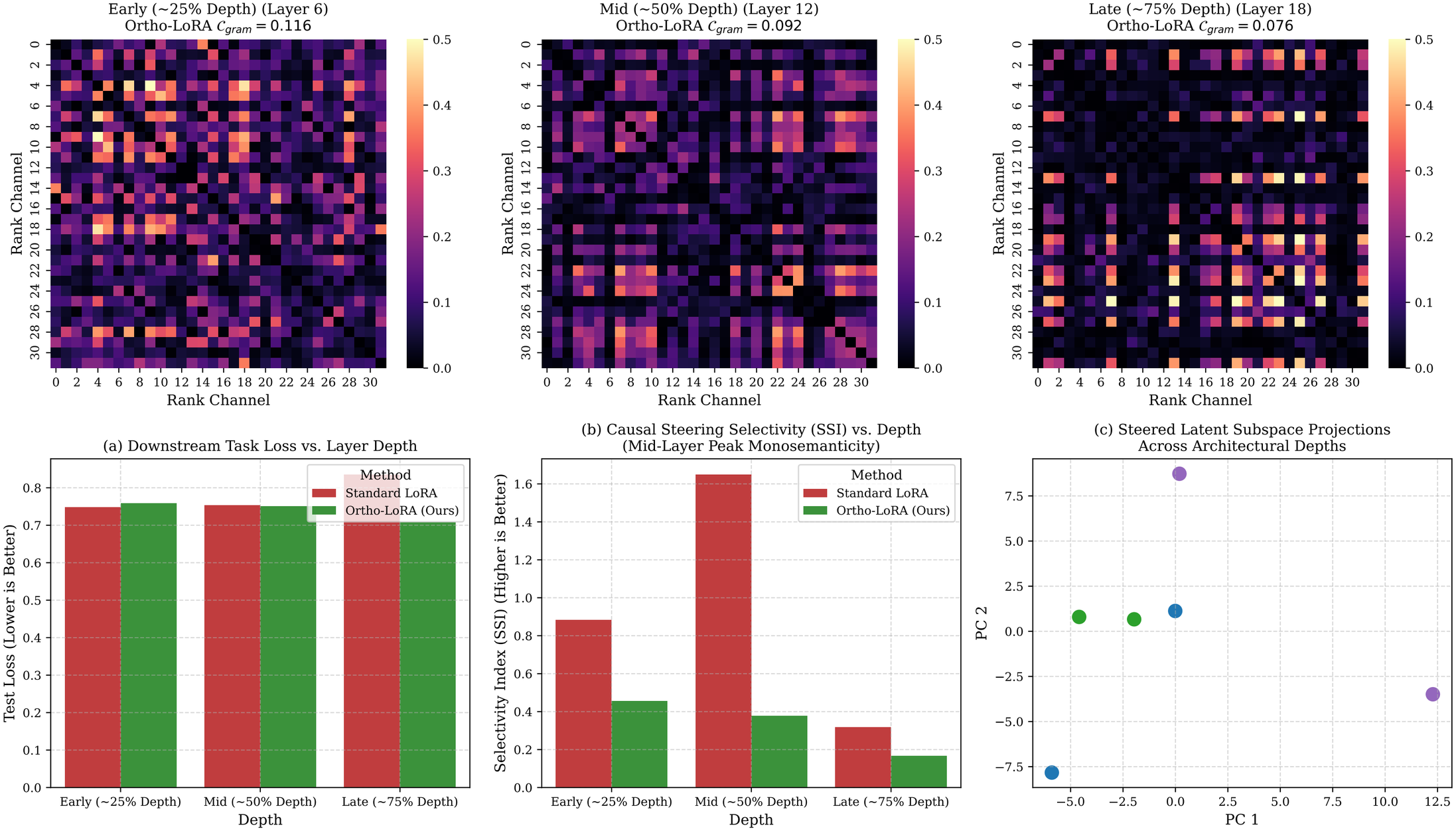

Early (~25% Depth) (Layer 6)
Ortho-LoRA $\mathcal{C}_{gram} = 0.116$
Mid (~50% Depth) (Layer 12)
Ortho-LoRA $\mathcal{C}_{gram} = 0.092$
Late (~75% Depth) (Layer 18)
Ortho-LoRA $\mathcal{C}_{gram} = 0.076$
Rank Channel
Rank Channel
(a) Downstream Task Loss vs. Layer Depth
Method
Standard LoRA
Ortho-LoRA (Ours)
Test Loss (Lower is Better)
Early (~25% Depth)
Mid (~50% Depth)
Late (~75% Depth)
Depth
(b) Causal Steering Selectivity (SSI) vs. Depth
(Mid-Layer Peak Monosemanticity)
Selectivity Index (SSI) (Higher is Better)
(c) Steered Latent Subspace Projections
Across Architectural Depths
PC 1
PC 2
Early (~25% Depth) (Standard)
Early (~25% Depth) (Ortho)
Mid (~50% Depth) (Standard)
Mid (~50% Depth) (Ortho)
Late (~75% Depth) (Standard)
Late (~75% Depth) (Ortho)